

INDIAN STATISTICAL INSTITUTE, KOLKATA
Assignment 6 , Second Semester 2024-25
Algebra 2 , M1
Date :

1. Let F be a field. Show that there are infinitely many irreducible polynomial in $F[x]$.
2. Show that $x^p - x$ has p distinct roots in $\mathbb{Z}/p\mathbb{Z}$.
3. Let k be a field and p be a prime such that $\text{char}(k) = p$. Show that all the roots $x^p - a$ are equal in any splitting field of $x^p - a$ over k . Conclude that if $x^p - a$ does not have any root in k then it is irreducible over k .
4. Let $k = \mathbb{F}_p$.
 - (a) Let L/k be a splitting field of $f(x) = x^p - x + a$, where $a \neq 0$. Show that all the roots of $f(x)$ are distinct in L .
 - (b) Show that $f(x) = f(x + 1)$.
 - (c) Show that if $\alpha \in L$ is a root of f then $\alpha + j$ is also a root of $f(x)$ for $j \in \mathbb{F}_p$. Conclude that f does not have any root in $\mathbb{F}_p$.
 - (d) Let g be any irreducible factor of f in $\mathbb{F}_p[x]$ and $\alpha \in L$ be any root of f in L . Then show that the sum of roots of g is $k\alpha + r$, where $k = \deg(g)$ and $r \in \mathbb{F}_p$, $k\alpha \neq 0$. Conclude that f is irreducible.
5. Let p be a prime and let $f(x) = \sum_{i=0}^{p-1} x^i$. Find out a splitting field K of $f(x)$ over $\mathbb{Q}$. Show that $[K : \mathbb{Q}] = p - 1$.
6. Let $f(x) = x^3 - 2$. Find out a splitting field K of f over $\mathbb{Q}$. Show that $[K : \mathbb{Q}] = 6$.
7. Construct a splitting field K of $x^4 - 2$ over $\mathbb{Q}$. Find out $[K : \mathbb{Q}]$.
8. Show the following :
 - (a) A field of cardinality p^k contains a subfield of order p^r iff $k|r$.
 - (b) Irreducible factors of the polynomial $x^q - x$ over $\mathbb{F}_p$ are the irreducible polynomials $\mathbb{F}_p[x]$ whose degrees divide r , here $q = p^r$.
9. Let $f(x) \in \mathbb{F}_p[x]$ be an irreducible polynomial of degree n , and let $q = p^n$. Show that $f(x) | x^q - x$. Conclude that $f(x)$ can not have repeated roots in any splitting field of $f(x)$ over $\mathbb{F}_p$.